

MAGNETOM

System

Troubleshooting Instructions

Cooling

Aera

Skyra

Avanto fit

Skyra fit

Prisma

Prisma fit

M7-000.840.18.08.02 | 06.21

3 Procedures

3.1 SEP

3.1.1 Drain procedure of the SEP (secondary water circuit)

i

The secondary water circuit contains distilled water with Natriumbicarbonat (NaHCO_3) (Label in the SEP cabinet)
Chilled water should be collected while draining.
This water should be refilled into the cooling circuit.
Or the cooling water must be disposed accordingly to the country specific environmental regulations and laws.
Avoid contact with eyes and skin.

3.1.1.1 Required tools

Filling adapter (part of system delivery)

Bucket

Water hose (part of system delivery)

3.1.1.2 Procedure

Switch off the compressor unit

Switch off circuit breaker EPC / F102





Fig. 7: SEP

(1) Main power switch

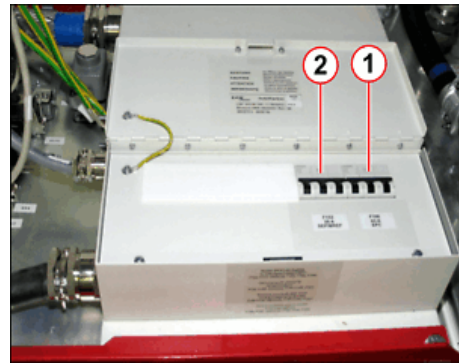


Fig. 8: Circuit breakers in EPC mains box

(1) F100 63A EPC

(2) F102 25A SEP/IFP - MREF

Drain the complete secondary water circuit of the SEP

Close the shutoff valves on top of the SEP  1/ Fig. 9

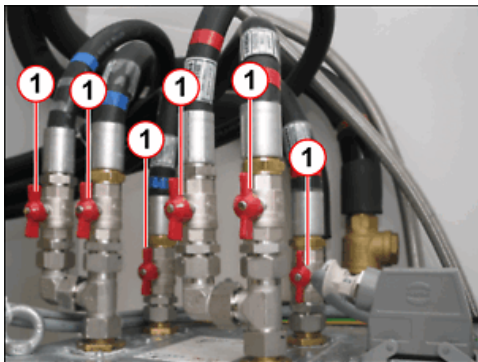


Fig. 9: SEP water connection

(1) Shutoff valve

Drain the secondary water circuit of the SEP for component exchange

Close the ball valves  Fig. 10,  1/ Fig. 11.



Fig. 10: Ball valve with strainer



Fig. 11: SEP

(1) Ball valve



Fig. 12: Expansion vessel

(1) Shut-off valve

Close the shut-off valve on the expansion vessel.

Connect the water hose to the drain and fill port of the SEP.

Run the water hose to a drain or a container.

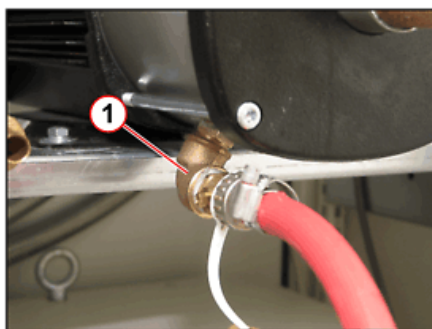


Fig. 13: Drain / fill port

(1) Drain / fill port secondary circuit

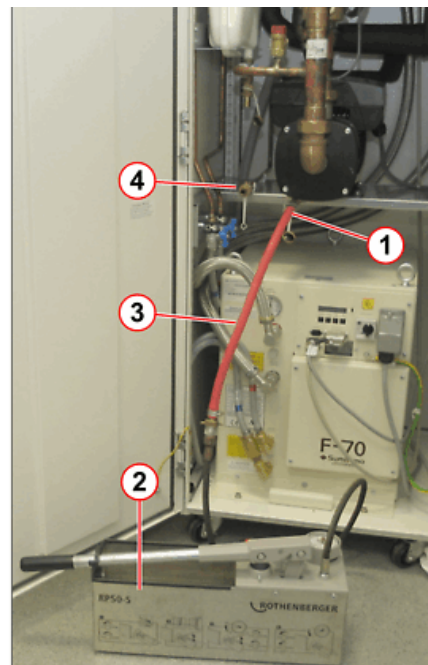


Fig. 14: SEP

- (1) Drain / fill port secondary circuit
- (2) Water pump
- (3) Water hose
- (4) Drain / fill port primary circuit



Fig. 15: Drain / fill port

Open the drain port using the cap of the drain valve.

Loosen the screw of the valve above the air bleeder a little. Air entering the system will help drain the SEP.

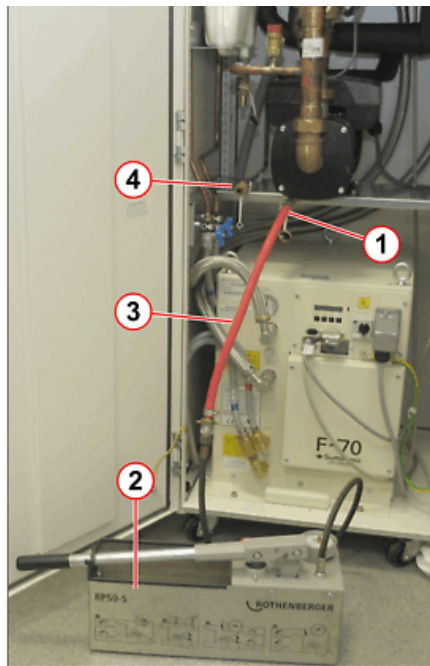


Fig. 16: SEP

- (1) Drain / fill port secondary circuit
- (2) Water pump
- (3) Water hose
- (4) Drain / fill port primary circuit

3.1.2 Filling the SEP

i

Do not use any connection to a drinking water system.
The secondary water circuit of the SEP needs to be filled with distilled water / deionized water.

3.1.2.1 Aids and Tools

Distilled water / de-ionized water

Water pump (manual water pump part # 8115136)

Filling adapter (part of system delivery)

Water hose (part of system delivery)

3.1.2.2 Procedure: Filling the water circuit

Switch off the compressor unit using the main power switch.

Switch off the water pump using Q3.



Fig. 17: SEP

(1) Main power switch

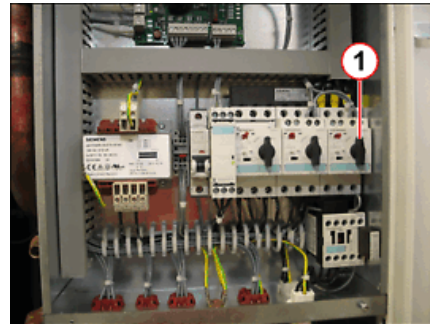


Fig. 18: SEP electronic cabinet

(1) Q3 water pump

Connect the water hose from the on-site water supply to the filling port of the SEP

or

connect the water hose from the water pump to the filling port of the SEP

i

Before connecting the hose to the SEP, the hose should be filled with water and not contain air.

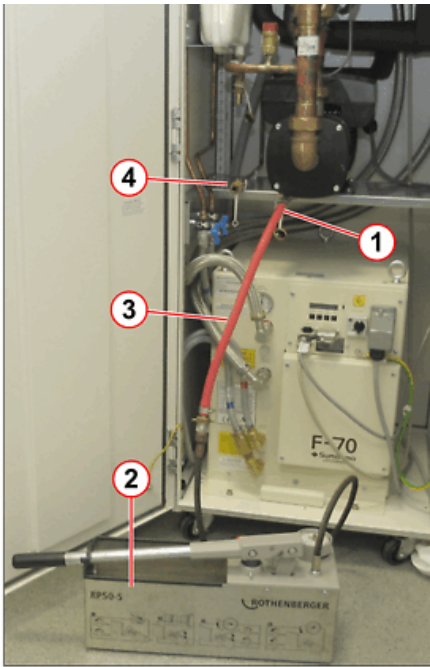


Fig. 19: SEP

- (1) Drain / fill port secondary circuit
- (2) Water pump
- (3) Water hose
- (4) Drain / fill port primary circuit

Open the drain/fill port valve using the back of the cap
Pump water into the circuit.
Close the valve when the pressure manometer indicates the correct pressure:

	SEP pressure manometer
Static pressure	1.0 ± 0.1 bar

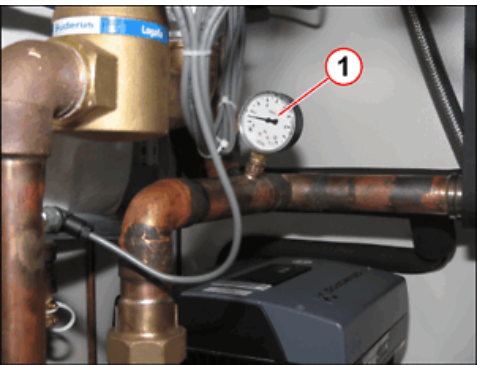


Fig. 21: SEP water pressure

- (1) Pressure manometer



Fig. 20: Drain / fill port



Fig. 22: Pressure manometer

Switch on the water pump Q3.

- Switch on the compressor unit.
- Check the system for leakages.
- Remove the filling hose.
- Close the filling port with the blanking plug.
- Check the fill pressure of the auto-refill device
 - » Refill it if necessary.

3.1.3 Filling the auto-refill device.

This description is valid for a SEP with an auto-refill device.

i

Do not use any connection to a drinking water system.
The auto refill device needs to be filled with distilled water / deionized water.

i

The auto-refill device maintains the secondary water return pressure constant at 1.0 bar.
Water is refilled automatically if water loss appears.
Prerequisite: The auto-refill device is filled with water and the pressure regulator is adjusted to 1.0 bar.

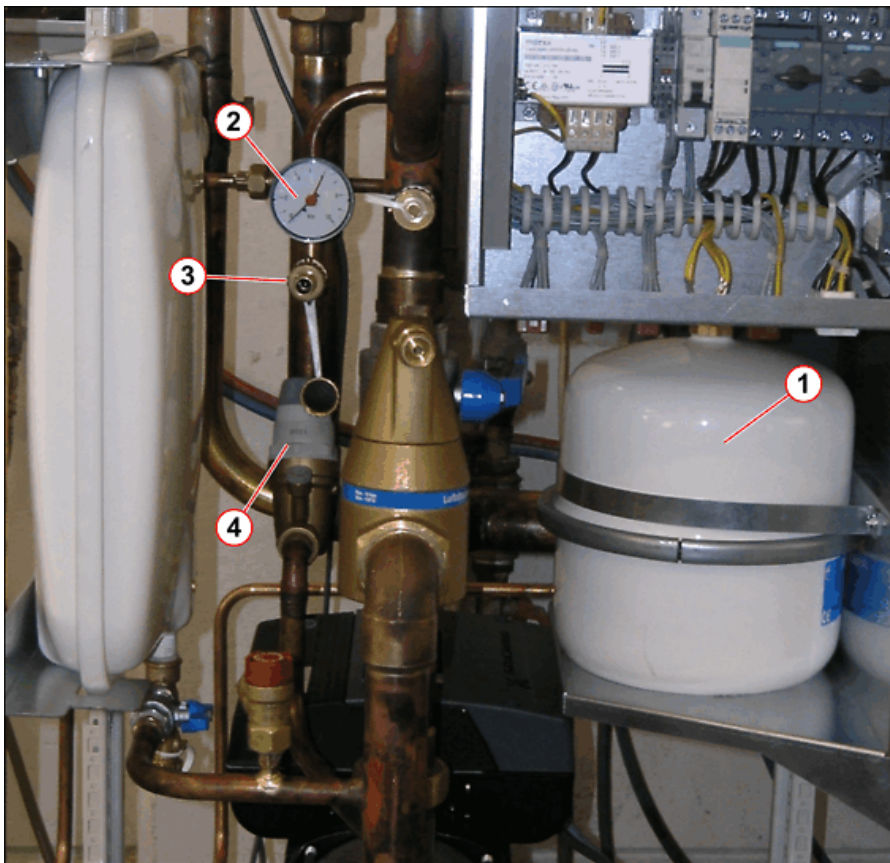


Fig. 23: Auto-refill device

- (1) Expansion vessel
- (2) Pressure manometer
- (3) Fill port of auto-refill device

(4) Pressure regulator

i

The expansion vessel should be checked at every service visit and should be filled to a **pressure of 6.0 - 0.5 bar**.

3.1.3.1 Aids and Tools

Distilled water / de-ionized water

Water pump (manual water pump part # 8115136)

Filling adapter (part of system delivery)

Water hose (part of system delivery)

3.1.3.2 Procedure

Connect a water hose to the fill port of the auto-refill device.  3/ Fig. 23

Open the fill port slowly.

Fill the auto-refill device to a **pressure 6.0 ± 0.5 bar**.

Check the pressure via the pressure manometer.  2/ Fig. 23

3.1.4 Cleaning the secondary water strainer

3.1.4.1 Required tools

50 mm open-end wrench

Needle nose pliers

Filling adapter (part of system delivery)

Water hose (part of system delivery)

Water pump (if no on-site water supply is available, manual water pump material number 8115136)

Bucket (cup)

3.1.4.2 Procedure

Switch off the compressor unit using the main power switch.  1/ Fig. 7

Switch off the water pump using Q3.  1/ Fig. 18



Fig. 24: SEP
(1)Main power switch

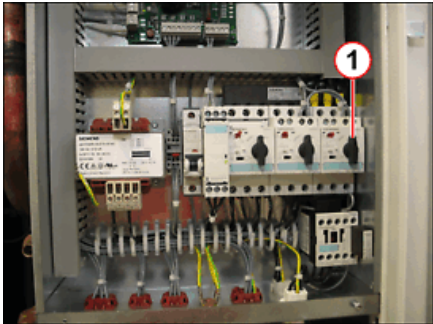


Fig. 25: SEP electronic cabinet
(1)Q3 water pump

Slowly close the ball valve.  Fig. 27



Fig. 26: SEP - ball valve open



Fig. 27: SEP - ball valve - closed (lever in vertical position)

Remove the valve cap slowly by using an open-end wrench.

i

The closed ball valve contains approximately 200 ml of cooling water. Use a cup to collect the water.

Use a cup to collect the water.  2/ Fig. 29

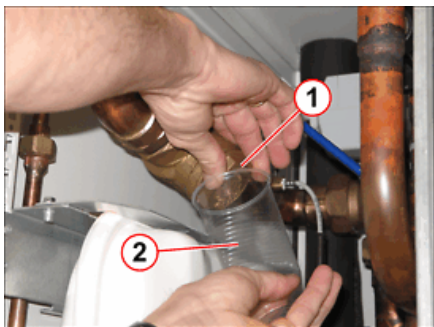
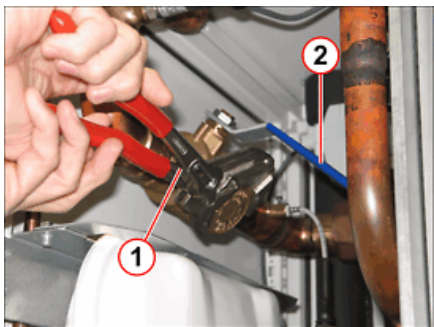


Fig. 28: SEP - ball valve with strainer

- (1) Open-end wrench
- (2) Ball valve - closed

Fig. 29: SEP - ball valve

- (1) Valve cap
- (2) Cup

Remove the circlip using needle nose pliers.  1/ Fig. 30

Remove the strainer.

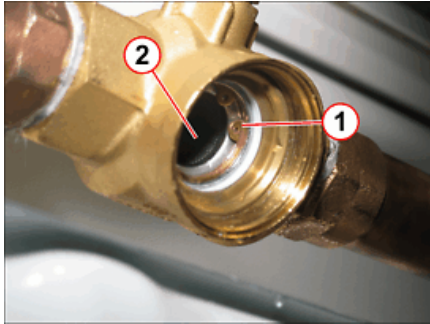


Fig. 30: SEP - strainer

- (1) Circlip
- (2) Strainer



Fig. 31: Strainer

Inspect the strainer for contamination and clean it by rinsing under tap water.

Open the drain/fill port valve using the back of the cap

Pump water into the circuit.

Close the valve when the pressure manometer indicates the correct pressure:

Reassemble the strainer

Re-install the strainer.

Mount the strainer with the circlip.

Check the O-ring of the valve cap.



Fig. 32: Ball valve cap

- (1) O-ring

Screw the valve cap into the ball valve.

Slowly open the ball valve.

Check for leaks on the valve cap.

Final steps

- Check the water pressure
- » Refill water if pressure < 1 bar

	SEP pressure manometer
Static pressure	1.0 - 0.5 bar

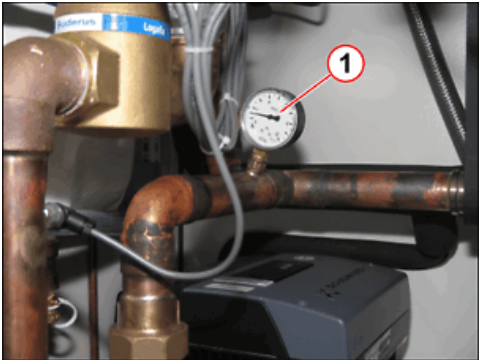


Fig. 33: SEP water pressure

(1)Pressure manometer



Fig. 34: Pressure manometer

- Switch on the water pump Q3
- Switch on the compressor unit.
- Check for leaks.

3.1.5 Cleaning the primary water strainer

Aids and tools

- 50mm or 41 mm open end wrench
- Needle nose pliers
- Cloth to absorb water
- Flash light
- Bucket

i

Cooling water containing ethylene glycol must be collected while draining. Return the cooling water to the water circuit during the next filling.
Avoid contact with eyes and skin.

- Switch off the MR system.
- Switch off the helium compressor
- Switch off the water pump using Q3
- Slowly close the valve  1/ Fig. 35.

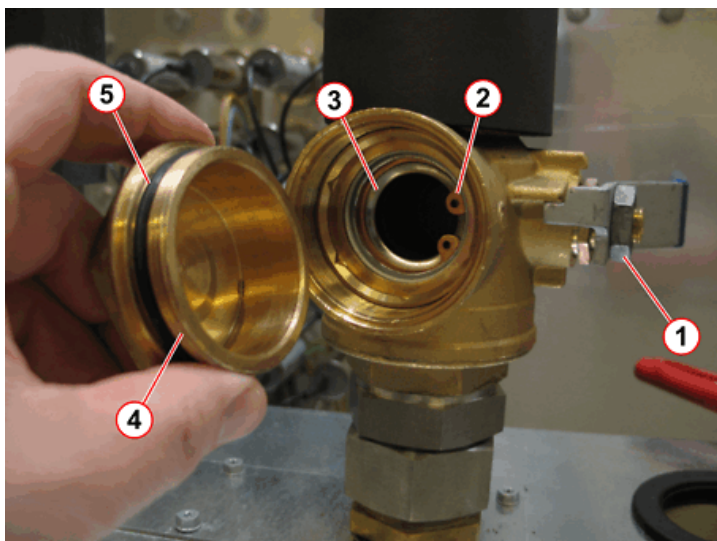


Fig. 35: SEP - valve with strainer

- (1) Handle - closed
- (2) Circlip
- (3) Strainer
- (4) Screw cap
- (5) O-ring

i

The closed valve contains approximately 200 mL cooling water. Use a bucket to collect the water.

Remove the screw cap 4/Fig. 35 slowly by using a open-end wrench.

Use a bucket to collect the water.

Use a cloth to absorb some water from the strainer hole.

Remove the circlip 2/Fig. 35 by using needle nose pliers.

Remove the strainer 3/Fig. 35.

Inspect the strainer for contamination and clean it by rinsing under tap water.



Fig. 36: Strainer

Reassemble the valve in reverse order.

- Install the strainer.
- Mount the strainer with the circlip.
- Check the O-ring  5/ Fig. 35 of the screw cap.
- Screw the screw cap into the shut-off valve.
- Slowly open the shut-off valve.
- Check for leaks on the screw cap.

Final steps

- Switch on the water pump using Q3
- Switch on the helium compressor
- Switch on the system.
- Open > Service Software GUI > Magnet & Cooling > **Cooling Status**
- Check the **Primary Water - Temperature and Flow**.
- If necessary, advise the customer to initiate corrective action immediately.

Secondary water circuit

- Check the water flow and pressure of the secondary water circuit.
- » If **SEP - Return Pressure** < **0.9 bar** (recommended pressure value for refilling), **refill water**.

3.1.6 Testing the expansion vessel

i

Before replacing the pump because of pressure problems, test the pressure of the expansion vessel.

3.1.6.1 Required Tools

- Air pump with auto-valve delivered with the system  Fig. 38.
- Measuring device: Manometer part #3055514  Fig. 37.
- Standard tool set.
- Water pump (manual water pump, part #8115136),  2/ Fig. 14.
- Water hoses  3/ Fig. 14.



Fig. 37: Measuring device



Fig. 38: Air pump



Fig. 39: Autovalve

3.1.6.2 Procedure

Switch off the compressor unit.

Switch off circuit breaker EPC / F102.

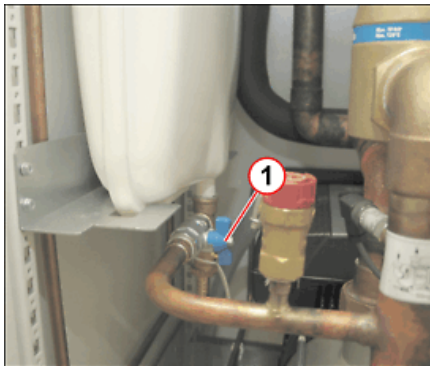


Fig. 40: Expansion vessel

(1) Shutoff valve

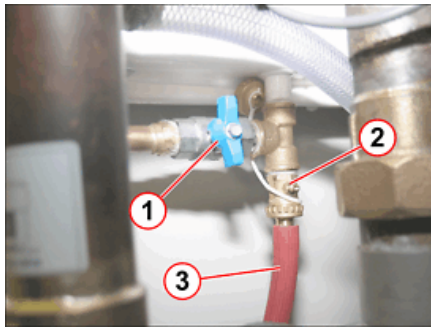


Fig. 41: Expansion vessel

- (1)Shutoff valve
- (2)Drain port
- (3)Water hose

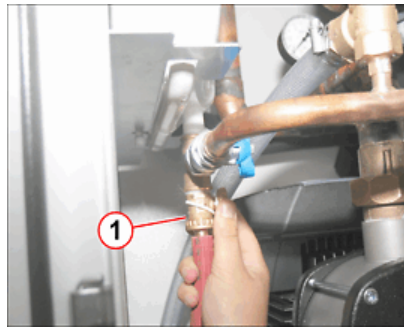


Fig. 42: Expansion vessel

- (1)Drain port

Close the shut-off valve  1/ Fig. 40.

Connect the water hose to the drain port  3/ Fig. 41.

Drain the expansion vessel.

» The volume of expansion vessel is about 8 l of water.

Remove the cap from the valve  Fig. 44.

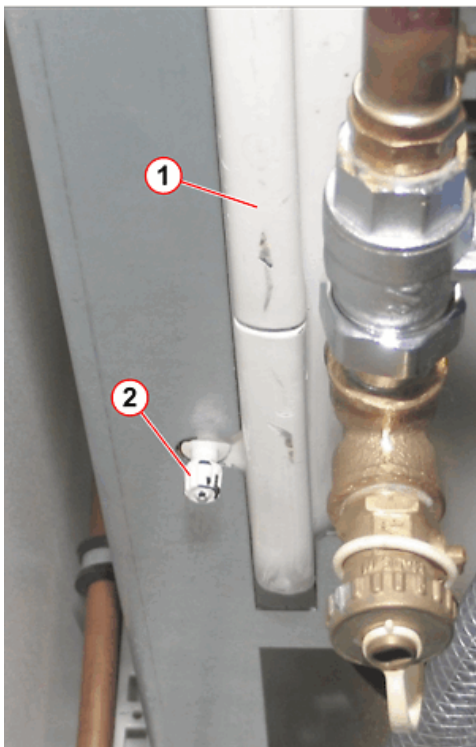


Fig. 43: Expansion vessel

- (1)Expansion vessel
- (2)Cap



Fig. 44: Cap

Connect the measuring device to the fill valve  Fig. 46,  1, 2/ Fig. 45.

Check the pressure.

» Pressure should be 0.5 bar.

If not, pressurize the vessel with the air pump.

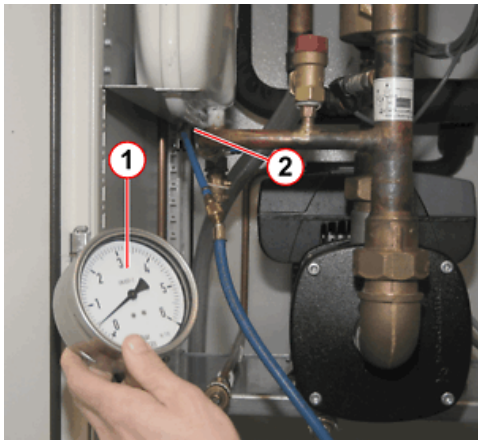


Fig. 45: Measuring device

(1) Measuring device

(2) Fill valve



Fig. 46: Fill valve

3.1.6.3 Pressurizing the expansion vessel

Connect the air pump to the fill valve  1, 2/ Fig. 48.

Pressurize the vessel to 0.5 bar.



Fig. 47: Air pump

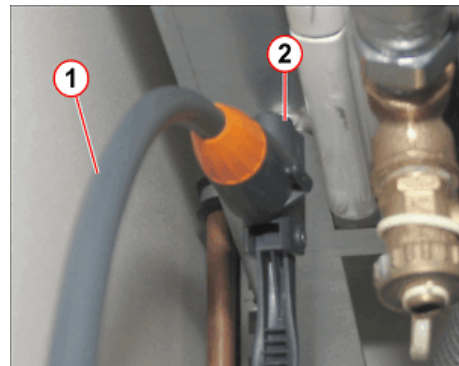


Fig. 48: Air pump

(1) Air pump

(2) Fill valve

3.1.6.4 Concluding Steps

- Undo the cap on the fill valve.
- Close the drain valve of the expansion vessel.
- Open the shut-off valve of the expansion vessel.
- Fill the secondary water circuit with water, refer to > Filling the SEP.
- Switch on F102.
 - » Water pump will start automatically.
- Start the compressor.
- Check the water pressure after the air is bled.

3.1.7 Conversion list of temperature sensors

Temperature (° C)	PT 100 (0.4 ohms/ ° C)	KTY (~15 ohms/ ° C)
	Location: ACS: R1, IFP: Temperature sensor	Location: SEP: B1, B2 (temperature sensor)
0	100 ohms	1,634 ohms
25	110 ohms	2,000 ohms +/- 10 ohms
10	104 ohms	1,774 ohms
20	108 ohms	1,922 ohms
30	112 ohms	2,078 ohms
40	116 ohms	2,040 ohms
50	120 ohms	2,410 ohms

3.2 Chiller

i

For detailed information about chiller topics refer to:
[KKT-chillersUSA](#)

i

The the auto-refill device should be checked at any service visit and should be filled to a pressure **6.0 ± 0.5 bar**.

3.2.1 Filling the chiller

i

Do not use any connection to a drinking water system.
Use the glycol mixture from KKT called KKT Protect 37

i

The water circuit from the chiller to the MR system needs to be filled with 35% to a maximum of 38% vol. ethylene glycol.
35% to 38% vol. ethylene glycol mixture corresponds to approx. -21 °C to -23 °C (-5.8 °F to -9.4 °F). Only ethylene glycol may be used. When starting up the chiller, the chiller start-up personnel should leave 30 liters of ethylene glycol mixture at the system. This reserve supply can be used for refills.
Use the mixture from KKT called KKT Protect 37.

i

Before connecting the water hose to the chiller, the hose should be filled with water and not contain air.

3.2.1.1 Aids and Tools

Water glycol mixture from KKT, called KKT Protect 37, material # 11344551)

Water pump (manual water pump material # 8115136)

Filling adapter

Water hose

3.2.1.2 Procedure

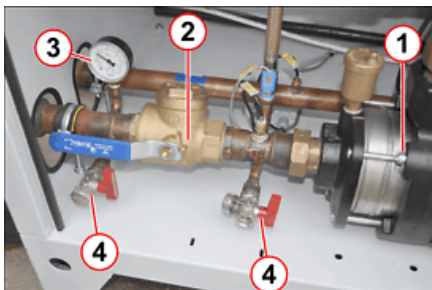


Fig. 49: Chiller

- (1) Water pump
- (2) Ball valve with strainer
- (3) Return water pressure manometer
- (4) Drain and fill port

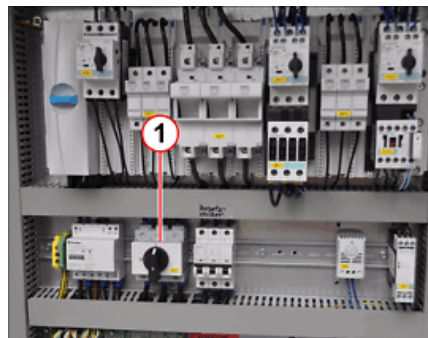


Fig. 50: Chiller electronic cabinet

- (1) Switch 4Q1

Switch off the helium compressor unit.

Switch off the chiller, using switch 4Q1.

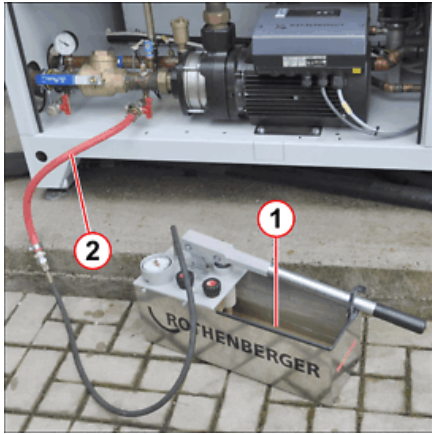


Fig. 51: Chiller

(1) Manual water pump

(2) Water hose

Fill the water pump with the mixture of ethylene glycol and water.

Fill the water hose with water before connecting it to the filling port.

Connect the water hose to the filling port.

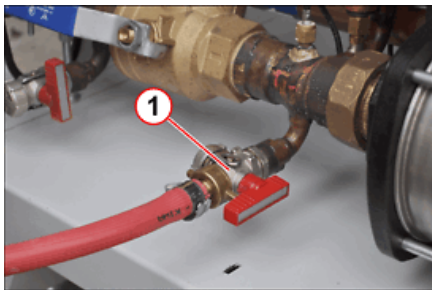


Fig. 52: Chiller

(1) Drain and fill port valve

Open the valve of the filling port.

Stop filling when the pressure manometer shows the correct value.



Fig. 53: Chiller

(1) Return water pressure manometer

Stop filling when the static pressure is 1.5 bar.

Check the pressure at the IFP manometer

Chiller **lower** than the IFP or on the **same level**

» static water pressure: 1.1 - 1.5bar.

Chiller **higher** than the IFP panel

» static water pressure 1.7 - 1.9bar.



Fig. 54: Chiller

Close the valve of the filling port.
 Disconnect the water hose.
 Close the filling port with the cap.

Check the fill pressure of the auto-refill device

» Refill it if necessary.

Switch on the chiller.

Switch on the helium compressor unit.

Check for water leaks.

3.2.2 Filling the Auto-Refill Device

These instructions are only for a chiller with an auto-refill device (second expansion tank).

i

The Chillers are equipped with Auto-refill device from serial number:

ECO133: 1153

ECO122: 1076

i

The auto-refill device maintains a constant water pressure in the water circuit..

Water is refilled automatically if water loss appears.

Prerequisite: The auto-refill device is filled with water / glycol mixture and the pressure regulator is adjusted to 0.6 bar.

Use the mixture from KKT called KKT Protect 37.



Fig. 55: Auto-Refill device

(1) Expansion vessel 25 l

(2) Fill/drain port of expansion vessel

- (3) Pressure manometer
- (4) Pressure regulator

3.2.2.1 Aids and Tools

Water glycol mixture from KKT, called KKT Protect 37, material # 11344551)

Water pump (manual water pump material # 8115136)

Filling adapter

Water hose

3.2.2.2 Procedure

Connect the water hose to the fill port of the auto-refill device .  2/Fig. 55

Open the fill port slowly.

Fill the auto-refill device to a **pressure 6.0 ± 0.5 bar.** .

» Check the pressure via pressure manometer .  3/Fig. 55

3.2.3 Filling at the ACS

i

Do not fill water / glycol mixture at the ACS with a running chiller water pump.

Switch off the chiller water pump for water / glycol mixture filling.

3.2.4 Testing the Ethylene Glycol Level

i

The specification is 35% vol. to a maximum of 38% vol. of ethylene glycol. Do not use propylene glycol or automotive antifreeze!

The chiller service engineer must perform a check of this concentration during chiller maintenance with a special measurement device.

Antifreeze testers are specified only for certain glycol brands.

3.2.4.1 Required Tools

A special antifreeze tester. Contact the chiller service department.

3.2.4.2 Procedure

Open the drain valve on the ACS or on the chiller and drain about 0.5 l ethylene glycol mixture into a bucket.

Using the temperature scale, check the ethylene glycol level with the ethylene glycol tester.

» -21 °C to -23 °C (-5.8 °F to -9.4 °F) corresponds to approx. 35% to 38% vol. of ethylene glycol water mixture.

Increase the ethylene glycol level if necessary.

i

Do not mix different types of glycol with one another, since this can result in flocculation and problems in the chiller.

If an incorrect type of glycol is added, the water circuit will need to be drained completely. Refill the water circuit with a 35% vol. to a maximum of 38% vol. of ethylene glycol water mixture.

Please contact the chiller start-up personnel.

If the chiller is operated without antifreeze, the chiller can be damaged.

3.2.5 Service Mode

i

All fault messages to MR System are disabled in Service mode.

3.2.5.1 Procedure

Switch to Service mode

Press the black reset button for at least 10 seconds.



Fig. 56: Chiller

(1)Reset button

Exit Service mode

Press the reset button for at least 10 seconds

» The automatic reset is activated at the latest after 2 hours.

3.2.6 Chiller Reset KSC 215

3.2.6.1 Procedure for KSC 215

KSC 215- Advanced traditional chiller(s) with logic for the power supply. As soon as the phase monitor recognizes it has overcome the phase lost, phase in-balance or over/under voltage condition, the chiller will restart. If the chiller has pending failures, it will not restart. The advantage of these chillers is the

ability to capture a fault in real-time via Siemens Simatic Display. The fault is captured and time stamped. To view older faults in the system we are able to enter the "General Fault" option.

When you encounter a KSC 215 in a failed state, you can do the following:

Simple Procedure

Observe the Simatic Display through the viewfinder of the chiller panel. A fault will display. (EXP: If faulted on a power issue, "Phase Sequence Relais" will appear.)

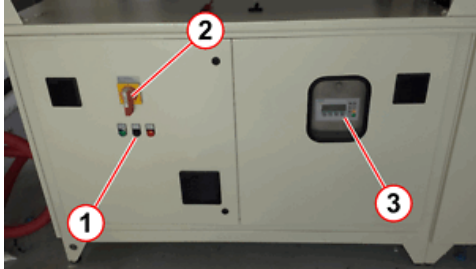


Fig. 57: Chiller KSC 215

- (1) Reset button
- (2) ON / OFF button
- (3) Simatic Display

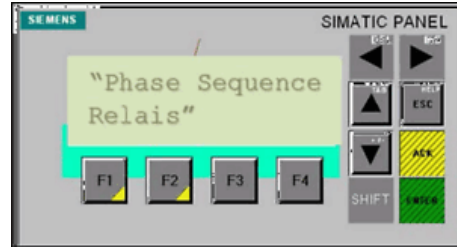


Fig. 58: Simatic display

Press the "Reset button"  1/ Fig. 57, chiller should run if no other fault condition is present.

If fault is still present, a complete shutdown of the chiller is recommended.  2/ Fig. 57

If fault is still present, contact the HSC and open a service call.

Advanced Procedure

After pressing the "Reset button" when the fault first appears, we can navigate through the controller (General Fault Option) to find past faults and when they occurred. (This step requires shutdown of the chiller to enter the electrical cabinet via key).

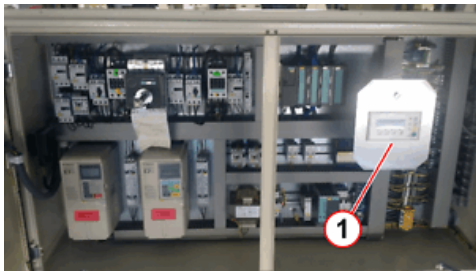


Fig. 59: Chiller Controller

- (1) Controller Display

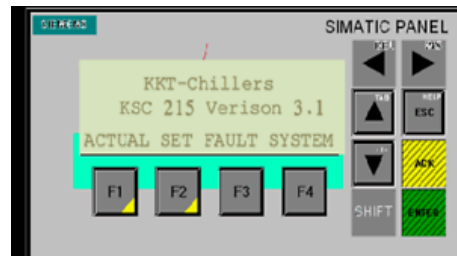


Fig. 60: Actual set fault system

Switch off the chiller.  2/ Fig. 57

Open the chiller cabinet.

Press F3, scroll through fault history.  2/ Fig. 61

Record the faults.

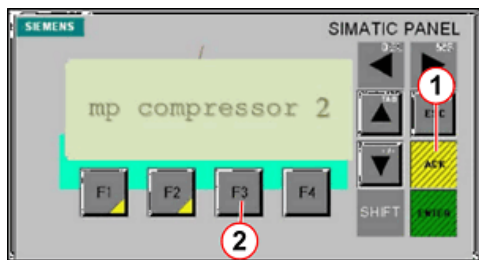


Fig. 61: mp compressor 2

- (1) ACK
- (2) F3

Press ACK, to acknowledge existing fault. 📎 1/ Fig. 61

Press "Reset button" on the door panel to clear the fault. 📎 1/ Fig. 57

Record the faults and forward to the HSC when opening a "Service Call". This will help diagnose if the issue is a power failure, etc.

Also, you may want to contact the HSC, and contact Tech Support for additional information. You may be able to receive information when the chiller initially faulted, especially if no fault is present. With the way the chiller communicates with the MR system via Data Cable, we are able to get a well defined failure on the system. (EXP: Compressor Failure, Pump Motor Protection, etc.)

3.2.7 Chiller Reset ECO

3.2.7.1 Procedure for the ECO series

ECO 122/133L- New generation of chiller(s) with logic for the power supply. As soon as the phase monitor recognizes it has overcome the phase lost, phase in-balance or over/under voltage condition, the chiller will restart. If the chiller senses pending failures, it will not restart. The advantages of these chillers are the ability to data log information when the chiller is operating.

When you encounter these chillers in a failed state, you can do the following:

Simple Procedure

View chiller faults at the IFP board 📎 1/ Fig. 62. The system mirrors any fault codes via fiber optic cable with a 1-8 LED.

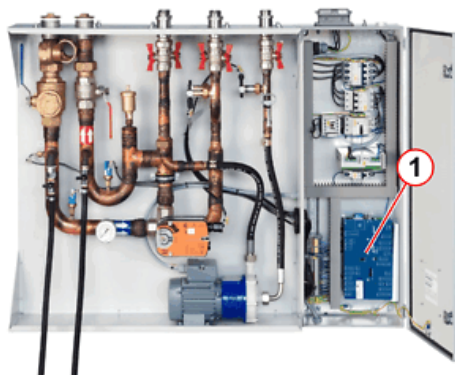


Fig. 62: IFP

- (1) IFP Board

Record the faults and forward it to HSC. Open a service call. This will help diagnose if the issue is a power failure, etc.

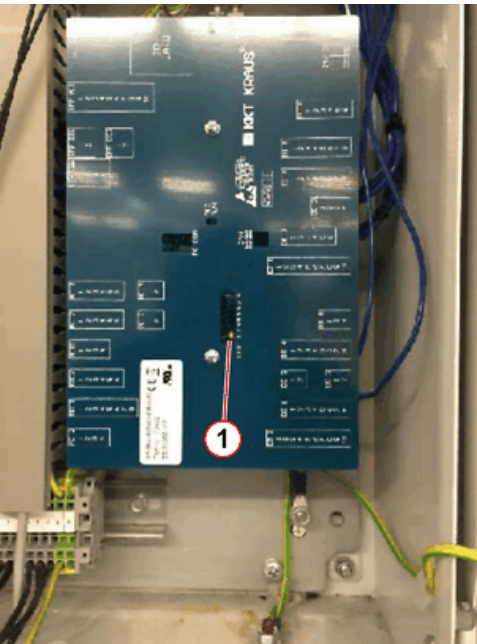


Fig. 63: IFB Board without fault
(1)Green LED's

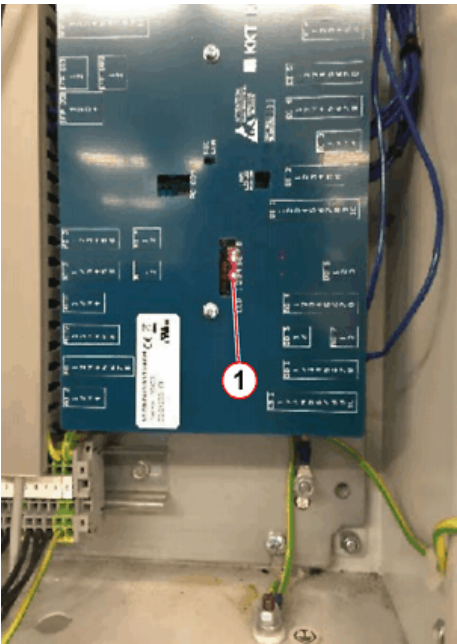


Fig. 64: IFB Board with fault
(1)Red LED's

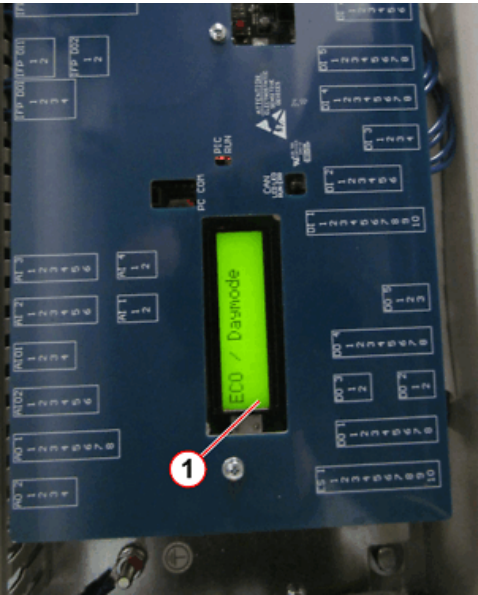


Fig. 65: IFB board V02 without fault
(1)Status display



Fig. 66: IFP board V02 with fault
(1)Status display

Advanced Procedure

Access the chiller control cabinet.
Open the door and record fault codes. 2/ Fig. 67

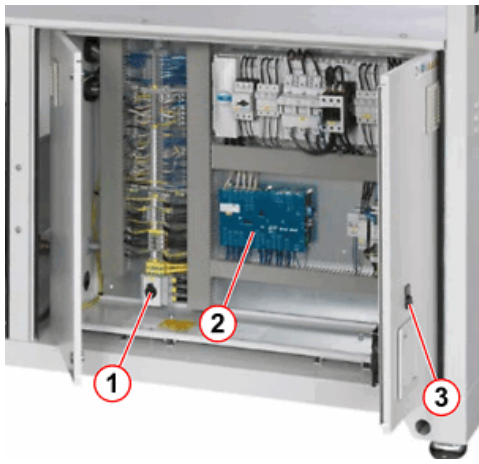


Fig. 67: ECO Chiller

- (1) Main switch
- (2) Chiller board
- (3) Reset button

Hold reset on the cabinet door for **three** seconds.  3/ Fig. 67

If cleared, chiller should not indicate any failures.

If faults are still present, switch off and switch on the chiller.  1/ Fig. 67

If fault is still present contact the HSC and open a service call..

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Also, you may want to contact the HSC, and contact Tech Support for additional information. You may be able to receive information when the chiller initially faulted, especially if no fault is present. With the way the chiller communicates with the MR system via fiber optic, we are able to get a well defined failure on the system. (EXP: Compressor Failure, Pump Motor Protection, etc.)

For additional support, you can send the chiller SD-Card files to Techsupport@kkt-chillersusa.com or mail to the KKT USA Office.  1/ Fig. 68

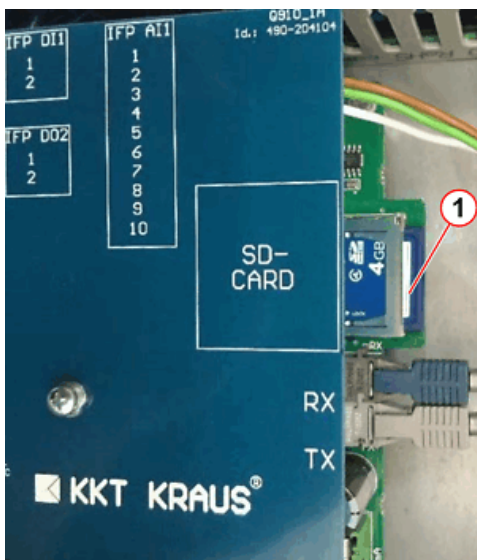


Fig. 68: Chiller board

- (1) SD card

3.2.8 Cleaning the strainer of the IFP and the Chiller

Aids and Tools

55mm open-end wrench

Needly nose pliers

Cloth to adsorb water

Bucket

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Cooling water containing ethylene glycol must be collected while draining. Return the cooling water to the water circuit during the next filling.
Avoid contact with eyes and skin.

Open > Service Software GUI > Magnet & Cooling > **Cooling Status**

Check the water flow and pressure of the water circuit.

Cleaning the strainer of the IFP

Switch off the system.

Switch off the helium compressor.

Switch off the chiller  1/ Fig. 69

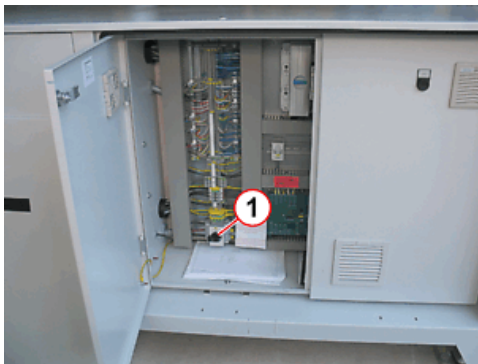


Fig. 69: Chiller - electrical cabinet

(1) ON / OFF switch



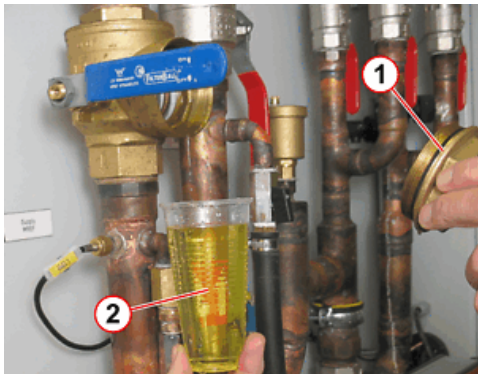
Fig. 70: IFP

(1) Valve with strainer; open

Slowly close the valve (1).

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The closed valve contains approximately 200 mL cooling water. Use a bucket to collect the water.



Use a bucket to collect the water.

Remove the screw cap (1) slowly by using an open-end wrench.

Use a cloth to absorb some water from the strainer hole.

Fig. 71: IFP - valve with strainer; closed

(1) Valve cap

(2) Cooling water - approximately 200 mL

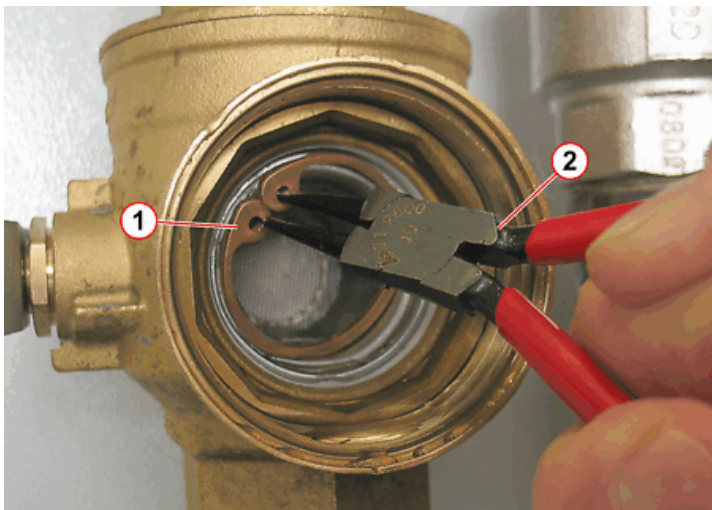


Fig. 72: Valve with strainer

(1) Circlip

(2) Needle nose pliers

Remove the circlip (1/Fig. 721) by using needle nose pliers.

Remove the strainer.

Inspect the strainer for contamination and clean it by rinsing under tap water.

Reassemble the valve in reverse order.

Install the strainer.

Mount the strainer with the circlip.

Check the O-ring of the screw cap.

Screw the screw cap into the shut-off valve.

Slowly open the shut-off valves.

Cleaning the strainer of the Chiller 1/ Fig. 73.

Slowly close the valve  1/Fig. 73(1)

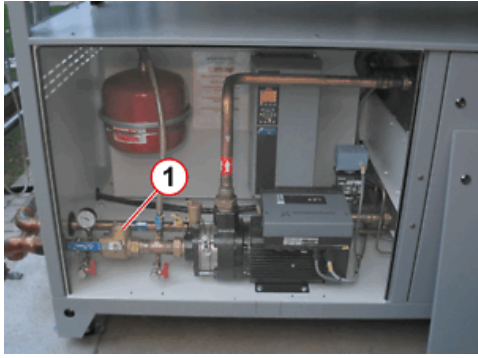


Fig. 73: Chiller - water cabinet

(1) Valve with strainer; open

Remove the screw cap slowly by using a open-end wrench.

Use a cloth to absorb some water from the strainer hole.

Remove the circlip by using needle nose pliers.

Remove the strainer.

Inspect the strainer for contamination and clean it by rinsing under tap water.

Reassemble the chiller valve in reverse order.

Install the strainer.

Mount the strainer with the circlip.

Check the O-ring of the screw cap.

Screw the screw cap into the shut-off valve.

Slowly open the shut-off valves.

Final steps

Switch on the chiller.

Check for leaks on the screw cap at the chiller..

Switch on the helium compressor.

Switch on the system.

Check for leaks on the screw cap at the IFP.

Open > Service Software GUI > Magnet & Cooling > **Cooling Status**

Check the pressure of the water circuit.

» If **IFP - water return** pressure < 1.0 bar (recommended pressure value for refilling), refill water (refer to "Filling the chiller")

» If **Chiller - fill / static pressure** < 0.7 bar (recommended pressure value for refilling), refill water (refer to "Filling the chiller").

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The pressure of the chiller and IFP are different depending on the location (above/below the MR system) where the chiller is installed.

